EMBL's European Bioinformatics Institute (EMBL-EBI)

The PRIDE Affinity-proteomics Archive (PRIDE-AP): making affinity proteomics datasets FAIR

EMBL-EBI

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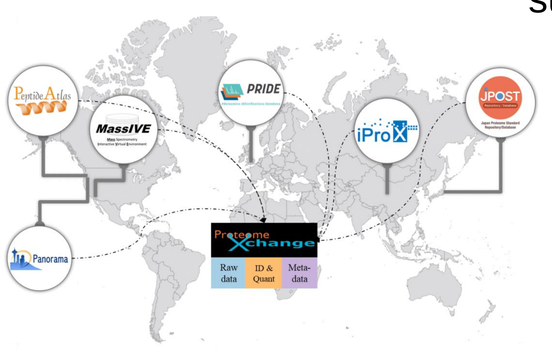
PRIDE contains ~47,000 MS datasets, from 80 countries, ~22,000 research groups

The PRIDE database at EMBL-EBI is the world-leading public data repository for mass spectrometry proteomics datasets. All proteomics approaches are supported. [1]

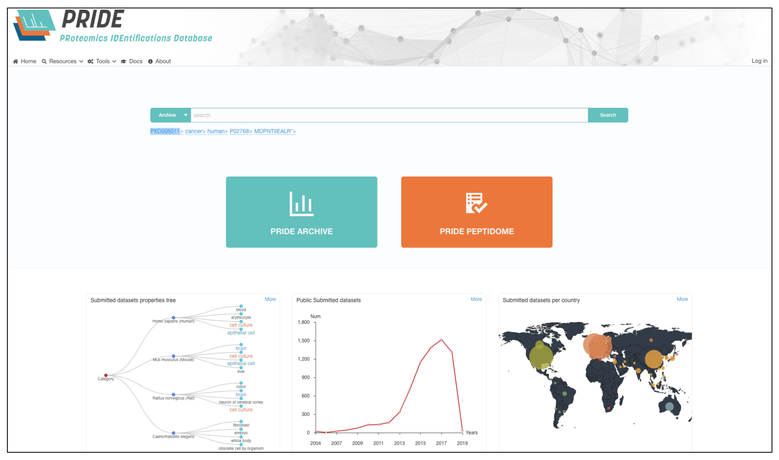


- ~69% datasets are publicly available (June 2024).
- PRIDE handles ~537 datasets/month.
- Automatic submission and validation system, with human intervention if required.

https://www.ebi.ac.uk/pride/

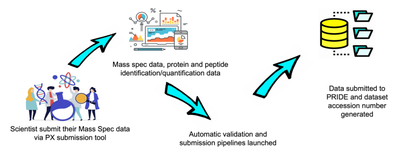
PRIDE is leading the ProteomeXchange consortium (https://www.proteomexchange.org), which aims to standardise data submission and dissemination of proteomics data worldwide. [2, 3]

Search and download proteomics data with the PRIDE web interface and API



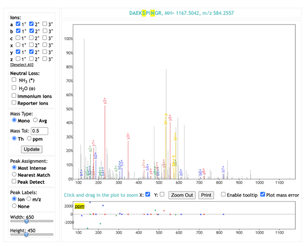
See PRIDE data policy at:
https://www.ebi.ac.uk/pride/markdownpage/datapolicy

Dataset submission and curation process



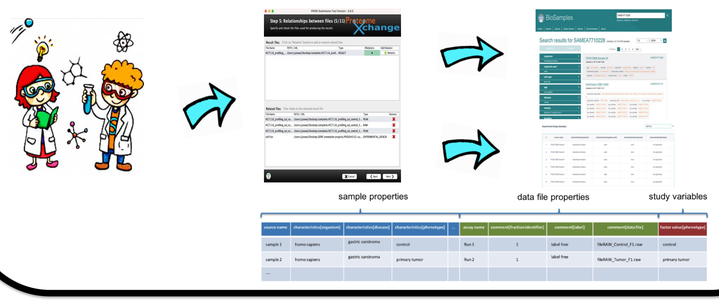
Proteomics data submission and arrival at PRIDE

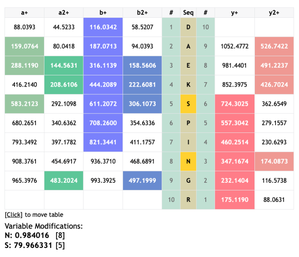
visualising the spectrum used to identify a given peptide as part of the sequence coverage of a particular protein [4]



PRIDE web interface provides functionality to assess the quality of each **submitted dataset**

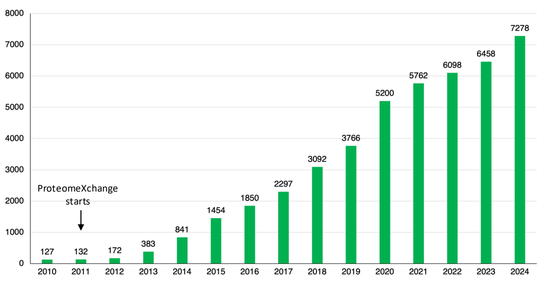
Submitters can now provide SDRF-Proteomics files to represent the experimental design and the relationship between the samples analyzed and the instrument raw files.
The samples included in the SDRF-Proteomics files are submitted to BioSamples getting each of them a unique accession number.
In addition, the PRIDE web interface represents the information contained in SDRF-Proteomics files in an 'Experimental Design' table, including all samples and data files.[5]



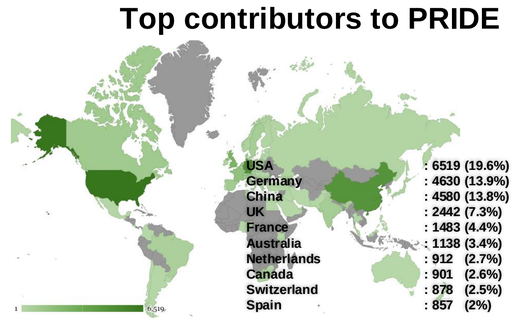


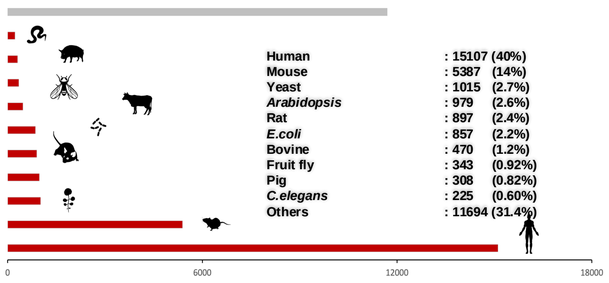
PRIDE MS statistics

Number of Submitted datasets per year

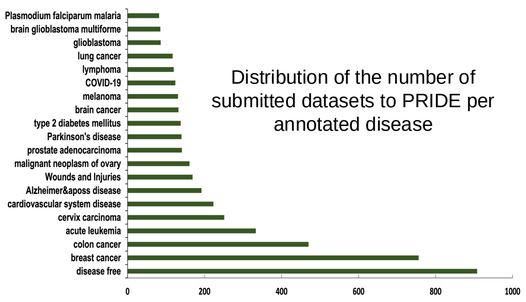


Top contributors to PRIDE





Top Species studied by at least 100 datasets



Distribution of the number of submitted datasets to PRIDE per annotated disease

The PRIDE Affinity-proteomics Archive (PRIDE-AP)

The PRIDE Affinity Proteomics (PRIDE-AP) Archive, is a new dedicated resource within PRIDE that aims to revolutionize affinity proteomics by centralizing openly accessible datasets generated using affinity proteomics platforms. PRIDE-AP Archive aims to establish an inclusive technology-agnostic repository representing all AP technologies. We have implemented a data submission workflow for the Olink (https://olink.com/) and SomaLogic AP (https://somalogic.com/) datasets. By adhering to the principles of Findability, Accessibility, Interoperability, and Reusability (FAIR), PRIDE-AP seeks to enhance collaboration, expedite research, and ensure the high-quality dissemination of affinity proteomics data.

PRIDE-AP Archive guidelines for the submission, validation and dissemination of affinity proteomics datasets

- A robust submission (using the PRIDE submission tool) and validation process to ensure datasets meet comprehensive metadata and file requirements. Each dataset is assigned a unique PAD identifier (e.g., PAD000001) for citation.
- Submitters must provide essential metadata, including title, description, sample and data analysis protocols, and information on data submitters and principal investigators.
- Currently supported technologies include Olink Proteomics (e.g. Explore & Target technologies) submissions with their specific 'RESULT' file type requirements (.parquet/.csv for Olink Proteomics).
- Additional files, such as experimental design documents and quality control reports, can also be included.
- The PRIDE-AP Archive infrastructure facilitates data submission, review, and public release, and the data remains unreleased to the public until publication.

Future plans, affects and impacts

We already have few datasets submitted and published to PRIDE-AP archive.

- We are working on SDRF for affinity proteomics data to enhance its impact by supporting it with comprehensive metadata. The structured metadata requirements and rigorous file validation processes ensure the creation of well-annotated, interoperable datasets.
- We are currently contacting research labs to submit their AP datasets to PRIDE.

The PRIDE-AP Archive will significantly advance affinity proteomics by adhering to the FAIR principles. It will centralize datasets from technologies like Olink Proteomics, facilitating collaboration, accelerating research, and promoting the dissemination of high-quality data. In the future, PRIDE-AP will evolve to support new technologies and expand its impact on the field.

https://github.com/PRIDE-Archive/pride-resources/blob/main/guidelines/pride-affinity-guidelines.md

References : 1. Perez-Riverol *et al.*, *NAR*, 2022, **2.** Vizcaíno *et al.*, *Nat Biotechnol*, 2014, **3.** Deutsch *et al.*, *NAR*, 2023, **4.** Perez-Riverol *et al.*, *NAR*, 2019, **5.** Chengxin Dai *et al.*, *Nat Comm*, 2021

Grants: 208391/Z/17/Z, 223745/Z/21/Z

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